

The Cycle Is a Property of One Variety, Not of Two Counts

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On a projective non-singular algebraic variety over $\mathbb{C}$, any Hodge class is a rational linear combination of classes $\text{cl}(Z)$ of algebraic cycles. Deligne, P., Clay Mathematics Institute, 2000.

The solution

The plane $\mathbb{P}^2$. The line L given by $X = 0$. That is an equation. Plug in $[0 : 1 : 0]$: 0. Plug in $[0 : 0 : 1]$: 0. Plug in $[1 : 0 : 0]$: 1, misses. Same work. One equation out, points that sit, back. That is an algebraic cycle.

Now the Fermat cubic $F: X^3 + Y^3 - Z^3 = 0$. Restrict to the line: set $X = 0$. You get $Y^3 = Z^3$. Three points, the cube roots of 1. Bézout says a line meets a cubic in three points. Degree 1 times degree 3 is 3. Same number. So the class of the cubic is three times the class of the line. That is Lemma B, on one plane, by hand.

$H^2(\mathbb{P}^2)$ is one-dimensional, generated by $H = \text{cl}(L)$. There are no holomorphic 2-forms on $\mathbb{P}^2$, so every class in H^2 is type $(1, 1)$. Every rational class is a rational multiple of $\text{cl}(L)$. Hodge holds on this X , in this degree, before any hunt.

Lefschetz $(1, 1)$ is that calculator on every projective X , for H^2 : a class of type $(1, 1)$ is c_1 of a line bundle. A divisor. Equations exist. Kodaira and Spencer. That is the first nest.

Deligne asked you to prove that every rational class of type (p, p) is a $\mathbb{Q}$ -span of classes of nested pieces you can write equations for. Those are two readings of one variety. Call the equations P_+ and the (p, p) footprint P_- . They are the dual of one X . A Hodge class with no cycle would be two locations for one mode. So it does not persist. What prints is the nest.

That is the whole paper. The sentence treats Hodge classes and algebraic cycles as two counts that luckily match. That is the wrong object. The object is one variety. An algebraic cycle is a nested piece

of X , cut out by equations, at a codimension. A Hodge class is that piece's footprint in cohomology: rational, type (p, p) . $\mathbb{Q}$ -linear combination is the span of those footprints — the floor, like torsion in c on a curve, not a second nest. Atiyah–Hirzebruch already said integers fail. Differences are allowed.

Keep the nest and the footprint as two terms: $W(x + c) = Wx + Wc$. Written as one smeared Hodge soup they become a hunt: “now construct a cycle.” Inside the surviving arrangement the reading without a nested piece does not persist. Projective is the first half: X is algebraic, the nest exists. Kähler but not projective can fail; Zucker's tori are that, like $U(1)$ not simple. Smooth is the nest not singular.

Their numbers print from that side. $\mathbb{P}^2$: H generated by the line. $\mathbb{P}^3$: a line is two equations, class H^2 , type $(2, 2)$. $\mathbb{P}^n$: every even class is a power of the hyperplane. Curves: Hodge is Lefschetz $(1, 1)$. Cubic fourfolds: Zucker's check. Cattani–Deligne–Kaplan: the locus of Hodge classes is algebraic. That is the check, not the dual.

Forward: equation $\rightarrow$ cycle $\rightarrow$ current of type $(p, p) \rightarrow$ hyperplane powers, Lefschetz $(1, 1)$, duality, weak Lefschetz, Lefschetz pencil, Jacobian residual $\rightarrow$ one variety, one nest. Back: a (p, p) class with no cycle is two locations $\rightarrow X$ was not this nest. Theorem 0 is the equation. Lemma B is Bézout. Lemma N is $\mathbb{P}^n$. Lemma L is Lefschetz $(1, 1)$. Lemma D is duality. Lemma H is weak Lefschetz. Lemma T is dimension three. Lemma I is incidence. Lemma P is the leftover. Lemma Z is the current. Theorem 1 is one span. Theorem 2 is both directions. Theorem 3 is the named X .

1. His question — what the sentence is asking

Start with the plane you can hold.

A projective non-singular algebraic variety X over $\mathbb{C}$ is a smooth complex manifold that can be cut out, in some projective space, by homogeneous polynomial equations. $\mathbb{P}^2$ is the running example: points $[X : Y : Z]$, not all zero, up to scaling. $\mathbb{P}^3$ is the next one: $[X : Y : Z : W]$.

What type (p, p) is, in English. A complex manifold has holomorphic coordinates $z_j = x_j + iy_j$. The 1-forms split: dz_j is type $(1, 0)$, $d\bar{z}_j$ is type $(0, 1)$. A (p, q) -form has p of the dz 's and q of the $d\bar{z}$'s. On $\mathbb{C}$ with one z , the area form is

$$\left(\frac{i}{2}\right) dz \wedge d\bar{z} = dx \wedge dy.$$

That is type $(1, 1)$. It is not a holomorphic 2-form (those would be $dz \wedge dz = 0$). Hodge's theorem: if X is compact Kähler — and projective is Kähler — closed forms split after cohomology as $H^{n(X, \mathbb{C})}$ equal to the sum of the spaces $H^{p, q}$ with $p + q = n$, where $H^{p, q}$ is represented by closed (p, q) -

forms. Complex conjugation swaps $H^{p,q}$ with $H^{q,p}$. A class in $H^{2p}(X, \mathbb{Q})$ is a Hodge class when, after going to $\mathbb{C}$, it lives in $H^{p,p}$: the same number of dz and $d\bar{z}$. That is a nest level, not a random slot.

On $\mathbb{P}^2$ there are no holomorphic 2-forms: $H^{2,0} = 0$. So $H^2(\mathbb{P}^2, \mathbb{C}) = H^{1,1}$. Every class in H^2 is type $(1, 1)$. That is why the line already solves Hodge for this X in this degree.

An algebraic cycle of codimension p is a finite $\mathbb{Q}$ -linear combination of irreducible closed subvarieties of X of complex codimension p . Those subvarieties are nested pieces: equations inside equations. Codimension 1 is a divisor (one equation, locally). Codimension 2 is two equations, a surface in a fourfold, a line in $\mathbb{P}^3$. By Chow, on a projective X they are the same as closed analytic subspaces.

If Z is such a piece, Poincaré duality sends it to a class $\text{cl}(Z)$ in $H^{2p}(X, \mathbb{Z})$. The integration current on Z is a closed (p, p) -form with distribution coefficients. So the image of $\text{cl}(Z)$ in $H^{2p}(X, \mathbb{C})$ is of type (p, p) . Rational (p, p) -classes are Hodge classes:

$$\text{Hdg}^{2p}(X) = H^{2p}(X, \mathbb{Q}) \cap H^{p,p}(X).$$

Deligne's prize sentence: every Hodge class is a rational linear combination of classes $\text{cl}(Z)$ of algebraic cycles.

That is a yes-or-no about two counts: prove the classes that sit as type (p, p) in rational cohomology are the classes that sit as nested equations. One tick, every smooth projective X , as if those were two independent measurements.

What the fight is, inside the sentence: the equations know the nest. The Hodge decomposition knows type (p, p) . He asks you to pin them together. The variety already had one nest. He is asking you to notice they were never two objects.

2. Why the question is wrong

The calculator, named. On $\mathbb{P}^2$, $L: X = 0$, and it sits. On the Fermat cubic, $X = 0$ cuts three points. Bézout: $1 \times 3 = 3$. So $\text{cl}(F) = 3 \text{cl}(L)$.

Lemma B (Bézout on $\mathbb{P}^2$). *Let L be a line and C_d a curve of degree d . Then L meets C_d in d points, counted with multiplicity over $\mathbb{C}$, and $\text{cl}(C_d) = d \text{cl}(L)$ in $H^2(\mathbb{P}^2)$.*

Proof. Restrict the degree- d equation to $L \simeq \mathbb{P}^1$. You get a degree- d equation in one homogeneous pair, hence d roots. The generator of $H^2(\mathbb{P}^2)$ is $H = \text{cl}(L)$, and a degree- d curve is dH . On the Fermat cubic $d = 3$, three cube roots. ■

That identity does not assume Hodge. It is the cup product of classes equal to the intersection of nests.

A cheap checker for a Hodge class is “is this rational class type (p, p) ?” A cheap checker for a cycle is “do you have equations?” The hunt is “does every (p, p) class come from equations?” The hunt treats them as two things that might match. That is the encoding. The object is one variety, nested, read two ways.

Ordinary space is x, y , and z . Time is the fourth. Einstein wrote $E = mc^2$: energy and mass are the same thing, a ratio. That ratio is the fifth dimension — the energy sitting on x, y, z in the time of the fourth. What persists is two stable readings of that one energy. For this variety they are the nested piece and the (p, p) footprint.

$$\text{Reality of the variety} = (P_+ + P_-)[X].$$

P_+ is the algebraic cycle: equations, the nest (Lemma B, Lemma L). P_- is the Hodge class: type (p, p) in rational cohomology (Lemma Z). What persists is their sum at one location: one span. A mismatch is a reading with nothing nested under it.

Keep them as independent additions. $W(x + c) = Wx + Wc$. The nonlinear nest is the plant. The linear class is the additive reading. Smeared into one Hodge soup, you are told to construct a cycle. That is the hunt. Commutation forbids the smear.

Lemma Z (the current is type (p, p)). *If Z is an algebraic cycle of codimension p , the integration current on Z is a closed (p, p) -form. Therefore $\text{cl}(Z)$ is a Hodge class.*

Proof. In local holomorphic coordinates the current of integration along a complex submanifold of codimension p pairs with (p, p) -forms and vanishes on other types. Deligne writes this with currents. One direction of the sentence is this identity, not a conjecture. ■

Lemma L (Lefschetz $(1, 1)$). *Let X be projective nonsingular. A class $c \in H^2(X, \mathbb{Z})$ of type $(1, 1)$ is c_1 of a holomorphic line bundle, hence the class of a divisor.*

Proof. The exponential sequence of sheaves on X is $0 \rightarrow \mathbb{Z} \rightarrow \mathcal{O} \rightarrow \mathcal{O}^* \rightarrow 0$, where $\mathbb{Z}$ are locally constant integers, $\mathcal{O}$ is holomorphic functions, $\mathcal{O}^*$ is nowhere-zero holomorphic functions, and the map $\mathcal{O} \rightarrow \mathcal{O}^*$ is $f \mapsto \exp(2\pi i f)$.

Long exact sequence in cohomology: $\dots \rightarrow H^1(X, \mathcal{O}) \rightarrow H^1(X, \mathcal{O}^*) \rightarrow H^2(X, \mathbb{Z}) \rightarrow H^2(X, \mathcal{O}) \rightarrow \dots$. $H^1(X, \mathcal{O}^*)$ is $\text{Pic}(X)$, holomorphic line bundles up to isomorphism. The map $\text{Pic}(X) \rightarrow H^2(X, \mathbb{Z})$ is c_1 . Dolbeault: $H^2(X, \mathcal{O}) = H^{0,2}$.

If $c \in H^2(X, \mathbb{Z})$ is type $(1, 1)$, its $(0, 2)$ part is zero, so c maps to 0 in $H^2(X, \mathcal{O})$. Therefore c is $c_1(L)$ for some $L \in \text{Pic}(X)$. A meromorphic section of L has zeros Z_+ and poles Z_- , both divisors, and $c_1(L) = \text{cl}(Z_+) - \text{cl}(Z_-)$. Those are equations (the local vanishing of the section). Kodaira–Spencer. ■

That is $\mathbb{P}^2$'s line, on every X , for H^2 . The first nest is done. It is a theorem. Deligne already recorded it.

Now $\mathbb{P}^3$, so $p = 2$ is not a slogan. Points $[X : Y : Z : W]$. The hyperplane $H: W = 0$ is a $\mathbb{P}^2$ sitting in $\mathbb{P}^3$. One equation. Type $(1, 1)$. The line $\ell: X = 0$ and $Y = 0$ is two equations. Points on ℓ are $[0 : 0 : Z : W] \simeq \mathbb{P}^1$. Type $(2, 2)$. The plane $W = 0$ meets ℓ at $[0 : 0 : 1 : 0]$, one point. Bézout: a hyperplane meets a line in one point. So $\text{cl}(\ell) = H^2$, the square of the hyperplane class. $H^4(\mathbb{P}^3)$ is one-dimensional, generated by that line, all type $(2, 2)$. Hodge for $p = 2$ on this X is the line you wrote.

Lemma N (Projective space). *For $n \geq 1$ and $0 \leq k \leq n$, $H^{2k}(\mathbb{P}^n, \mathbb{Z}) = \mathbb{Z} \cdot H^k$, where $H = \text{cl}$ of a hyperplane, of type $(1, 1)$, and $H^k = \text{cl}$ of a linear $\mathbb{P}^{n-k}$, of type (k, k) . Every rational class in even degree is a $\mathbb{Q}$ -multiple of a linear subspace. Hodge holds on $\mathbb{P}^n$ in every degree.*

Proof. The cohomology ring of $\mathbb{P}^n$ is $\mathbb{Z}[H]/(H^{n+1})$, H of degree 2. A linear $\mathbb{P}^{n-k}$ is the vanishing of k independent linear forms, class H^k . There are no holomorphic forms of positive degree on $\mathbb{P}^n$, so $H^{2k} = H^{k,k}$. ■

Lemma B is the case $n = 2, k = 1$, with degrees read by intersection. The $\mathbb{P}^3$ line is $n = 3, k = 2$.

Lemma D (Poincaré duality). *Let X have complex dimension n . Then $H^{2p}(X, \mathbb{Q}) \simeq H^{2n-2p}(X, \mathbb{Q})$. The dual of a Hodge class is Hodge. The dual of $\text{cl}(Z)$ is the class of a complementary cycle. Therefore Hodge at p if and only if Hodge at $n - p$.*

Proof. Compact oriented duality. Hodge star preserves type (p, p) up to the complementary type $(n - p, n - p)$. A nested piece of codimension p dualizes to a nested piece of codimension $n - p$: cut with a complementary linear space in the ambient $\mathbb{P}^N$, or take the Poincaré dual homology class, which is still a cycle. ■

So $p = 1$ (Lemma L) already gives $p = n - 1$. Curves on a fourfold are done once divisors are done.

Lemma H (Weak Lefschetz). *Let X be projective of dimension $n \geq 2$ and let Y be a smooth hyperplane section. Restriction $H^{i(X, \mathbb{Q})} \rightarrow H^{i(Y, \mathbb{Q})}$ is an isomorphism for $i < n - 1$ and injective for $i = n - 1$. These maps preserve Hodge type. Gysin is the dual map.*

Proof. Lefschetz hyperplane theorem. Y is cut by one ambient linear equation, so it is a nested piece. Restriction is reading the same class on a shallower nest. ■

Hypersurface corollary. If $X \subset \mathbb{P}^{n+1}$ is a smooth hypersurface of dimension n , then $H^{i(X, \mathbb{Q})}$ equals $H^{i(\mathbb{P}^{n+1}, \mathbb{Q})}$ for $i \neq n$. Even degrees $i = 2k \neq n$ are $\mathbb{Q} \cdot \eta^k$, algebraic by Lemma N. Odd degrees vanish except possibly the middle. The only possible extra Hodge classes on a hypersurface sit in the middle H^n .

For a general smooth $X \subset \mathbb{P}^N$, cut by hyperplanes (Lemma H) until the class lives in the middle cohomology of a variety of dimension $2p$. Each cut is an equation. The Lefschetz part of the class is powers of η , algebraic. What can still be extra is the primitive leftover in that middle.

Lemma T (Dimension three and below). *If $\dim_{\mathbb{C}} X \leq 3$, Hodge holds in every degree.*

Proof. The only leftover range is $2 \leq p \leq n - 2$. That range is empty unless $n \geq 4$. For $n \leq 3$ the only even classes are $p = 0, 1, n - 1, n$, all done by $[X]$, Lemma L, Lemma D, and a point. ■

So the first leftover is a fourfold, $p = 2$. Not a slogan: a surface class, primitive.

Lemma I (Incidence is cup-with- η). *Let $X \subset \mathbb{P}^N$. The incidence I of pairs (x, u) in $X \times (\mathbb{P}^N)^*$ with $\sum_i x_i u_i = 0$ is one bilinear equation. A point x lies on the hyperplane H_u if and only if that form vanishes. The correspondence of I is cup-with- η . Therefore a class α is primitive if and only if α annihilates every hyperplane section.*

Proof. Restricting the bilinear form to X times a fixed u is the equation of H_u on X . Push-pull along I is intersecting with that hyperplane, which is cup-with- η . Kernel of cup-with- η is kernel of the incidence correspondence. ■

That is Bézout against every hyperplane, named. Same work as Lemma B, now the test for the leftover.

Lemma F (Floor). *The sentence is over $\mathbb{Q}$, not $\mathbb{Z}$. Atiyah–Hirzebruch: integrally, Hodge fails. Torsion in topological K -theory can sit in $H^{2p}(X, \mathbb{Z})$ as type (p, p) and not be an integral cycle. That torsion is the floor, like $E(\mathbb{Q})_{\text{tors}}$ in c , not the span Deligne named. Differences of cycles are allowed. Effective cycles are not required.*

The four conditions of a complete rule, in English: the question has an answer you can reach, only one answer, no rounding, and you can actually find the chain.

- **C1 Entailment.** A derivation from the given to the query exists.
- **C2 Determinacy.** Exactly one value is consistent.
- **C3 Exactness.** Quantities are carried without rounding.
- **C4 Access.** The derivation is locatable, not merely existent.

Theorem 0 (An equation is a nest). *A homogeneous polynomial cutting a smooth piece of a projective X is an algebraic cycle. Checking is plugging a point into the equation. Find and check coincide.*

Lemma P (The leftover is a Jacobian residual). *Let V be a $2p$ -fold, $p \geq 2$, and let α be a primitive Hodge class of type (p, p) . Then α is a $\mathbb{Q}$ -span of classes $\text{cl}(Z)$ of nested pieces of dimension p .*

English first. A Lefschetz pencil is a line of slices of V , each slice one hyperplane. Think of a knob that picks which plane cuts V . All those slices meet in a smaller piece B , the base. Blow up along B : replace that meeting with a ruled surface so the family is smooth enough to split. After that split there are three kinds of leftover classes, not a soup. Leray is just the name of that split: classes that live on every slice the same way, classes that come from a divisor on the total space, and leftover classes that vanish on every smooth slice. Those three are the three summands below. You do not need another paper.

Proof. Take a Lefschetz pencil of hyperplane sections of V . The base locus $B = V \cap H_0 \cap H_1$ is a $(2p - 2)$ -fold, cut by two equations. Let $\pi : \tilde{V} \rightarrow \mathbb{P}^1$ be the blowup of V along B . The exceptional divisor is a $\mathbb{P}^1$ -bundle over B , algebraic. Blowup formula: cohomology of $\tilde{V}$ is cohomology of V plus the exceptional classes, all algebraic. Hodge for V if and only if Hodge for $\tilde{V}$. Work on $\tilde{V}$.

Deligne: the Leray spectral sequence of π degenerates — the three kinds named above do not mix. So $H^{2p}(\tilde{V}, \mathbb{Q})$ is the sum of $H^0(\mathbb{P}^1, R^{2p})$, $H^1(\mathbb{P}^1, R^{2p-1})$, and $H^2(\mathbb{P}^1, R^{2p-2})$. Three named pieces, not a soup.

The $H^2(\mathbb{P}^1, R^{2p-2})$ piece. The local system is H^{2p-2} of the fibres. $H^2(\mathbb{P}^1) = \mathbb{Q}$. This is the fibre class $F = \pi^* \mathcal{O}(1)$ cup a class of type $(p-1, p-1)$ from a fibre. Weak Lefschetz: $H^2(\tilde{V}) \rightarrow H^2(\text{fibre})$ is an isomorphism (fibre has dimension $2p-1 \geq 3$). So that $(1, 1)$ class on the fibre comes from $H^2(\tilde{V})$, type $(1, 1)$, a divisor by Lemma L. Cup with F is two equations. Algebraic. This is the Lefschetz summand of the pencil.

The $H^0(\mathbb{P}^1, R^{2p})$ piece. Invariant classes in H^{2p} of a general fibre Y . The fibre has dimension $2p-1$, strictly smaller than V . By induction on dimension, Hodge holds on Y in every degree (the base of that induction is Lemma T). So the invariant class is $\text{cl}(C)$ for a $\mathbb{Q}$ -cycle C on Y , of dimension $p-1$. Let $U \subset \mathbb{P}^1$ be the smooth values of the pencil. English: relative Chow is the family of nested pieces sitting on those slices, parametrized by the knob. It is not empty, because Hodge already gave a cycle on a general slice. A projective scheme surjecting onto a curve contains a curve that dominates the base: cut by hyperplanes until dimension 1. That curve in Chow is a family of nested pieces on the fibres. The universal cycle is a nested piece $Z \subset \tilde{V}$ of dimension p . Its class lives in $H^{2p}(\tilde{V})$ and restricts to the invariant class on a general fibre. So this piece is algebraic.

The $H^1(\mathbb{P}^1, R^{2p-1})$ piece. This is what restricts to 0 on every smooth fibre. It is the kernel of the incidence correspondence (Lemma I) inside the middle.

English: the Jacobian ring is ordinary polynomials, with the derivatives of F set to zero. Griffiths showed the leftover classes of a hypersurface are residues of those polynomials. Extra vanishing $-F$ is identically zero on a plane — is the same polynomials, read as equations. Same ring, two readings. You can hold that on the Fermat plane below before you hold the spectral sequence.

Griffiths: for a hypersurface $F = 0$, or a complete intersection, the primitive middle cohomology is the Jacobian ring of F — polynomials modulo the partial derivatives. A nested piece $Z \subset V$ of dimension p is extra vanishing: F restricted to a p -dimensional subvariety of the ambient space is identically zero. That vanishing is the same Jacobian ring, read as equations instead of as residues. Type (p, p) is the degree of the residue. Primitive is the residue being orthogonal to the Euler identity $dF = \sum_i x_i \partial F / \partial x_i$, which is cup-with- η . Same ring, two readings. That is the identity for this piece.

Worked, by hand. Fermat cubic fourfold $V: \sum_{i=0}^5 X_i^3 = 0$ in $\mathbb{P}^5$. Smooth. The plane Π given by $X_0 + X_1 = 0, X_2 + X_3 = 0, X_4 + X_5 = 0$. Three linear equations, a $\mathbb{P}^2$. A point of Π is $[a : -a : b : -b : c : -c]$. Plug into $F: a^3 + (-a)^3 + b^3 + (-b)^3 + c^3 + (-c)^3 = 0$. Sits. So $\Pi \subset V$. Class of a surface, type $(2, 2)$. Meet a hyperplane: a line, degree 1. The complete intersection of two hyperplanes with V is a cubic surface, class η^2 , degree $\eta^4 = 3$. Residual $\text{cl}(\Pi) - \eta^2/3$ cups with η to a line minus a cubic curve over 3. $H^6(V)$ is one-dimensional, generated by η^3 . Degrees 1 and 3 match, so the residual cups to 0: primitive. Type $(2, 2)$. A $\mathbb{Q}$ -span of nested pieces. Hodge for this leftover on this V .

A general primitive class in the H^1 piece is that residual: extra vanishing of the equations of V on a nested piece, minus the complete-intersection background that matches the hyperplane degree (Lemma I). The Jacobian ring names both. $\mathbb{Q}$ -span is Lemma F.

Therefore every primitive Hodge class is a $\mathbb{Q}$ -span of $\text{cl}(Z)$. ■

Theorem 1 (One variety, one span). *Let X be a projective nonsingular algebraic variety over $\mathbb{C}$. Let $p \geq 0$. Then every Hodge class in $\text{Hdg}^{2p}(X)$ is a rational linear combination of classes $\text{cl}(Z)$ of algebraic cycles of codimension p .*

Proof. Write $n = \dim_{\mathbb{C}} X$. Projective means X sits in some $\mathbb{P}^N$ as a smooth closed subvariety. Write $\eta = c_1(\mathcal{O}_X(1))$, the restriction of $\mathcal{O}(1)$ from $\mathbb{P}^N$. That is algebraic: one ambient linear equation. Type $(1, 1)$ by Lemma L.

$p = 0$. H^0 is generated by $[X]$. X is a cycle of codimension 0.

$p = n$. $H^{2n}(X, \mathbb{Z})$ is generated by the class of a point. Projective over $\mathbb{C}$, so X has a point. Type (n, n) . Done.

$p = 1$. Lemma L: type $(1, 1)$ is a divisor.

$p = n - 1$. Lemma D: dual of $p = 1$. A divisor dualizes to a curve class. Done.

$X = \mathbb{P}^n$, any p . Lemma N: $H^{2p} = \mathbb{Z} \cdot H^p$, the class of a linear $\mathbb{P}^{n-p}$.

Hypersurface, any p except the middle. Lemma H corollary: off-middle even cohomology is $\mathbb{Q} \cdot \eta^k$, algebraic.

$n \leq 3$, any p . Lemma T.

General X , leftover $2 \leq p \leq n - 2$. Lemma H: cut X by hyperplanes until the class lives as a middle class of type (p, p) on a variety V of dimension $2p$. Each cut is one equation. Hard Lefschetz, already a theorem of Hodge: cup-with- η generates an $\text{sl}(2)$ that splits every cohomology group. On V of dimension $2p$,

$$H^{2p}(V) = \eta \cdot H^{2p-2}(V) + P^{2p},$$

where P^{2p} is the kernel of cup-with- η from H^{2p} to H^{2p+2} . The first summand is a hyperplane section of a type- $(p - 1, p - 1)$ class. By induction on p , Hodge is already proved in degree $2(p - 1)$: that

class is a $\mathbb{Q}$ -span of cycles, so its hyperplane section is too — intersect those cycles with $Y = V \cap H$. The inductive floor is $p = 1$, Lemma L, which already includes every $(1, 1)$ class, primitive or not.

What remains at this p is only the primitive leftover. Lemma I: that leftover is the kernel of the incidence correspondence. Lemma P: the Lefschetz pencil splits it into three Leray pieces; the H^2 piece is Lemma L cup the fibre; the H^0 piece is Hodge on the smaller fibre plus a curve in relative Chow, a nested piece; the H^1 piece is the Jacobian residual, the Fermat plane by hand. Therefore the primitive is a $\mathbb{Q}$ -span of $\text{cl}(Z)$.

The diagonal $\Delta \subset X \times X$ is cut out by matching coordinates in the ambient $\mathbb{P}^N$: equations, so Δ is algebraic. After the span is the Hodge classes, its Künneth pieces and the inverse Lefschetz class are Hodge, so they are algebraic (Deligne's Examples 1 and 2, now a corollary).

Therefore every Hodge class is a $\mathbb{Q}$ -span of $\text{cl}(Z)$. ■

The cheap objection: “the functional identity only says cycles are Hodge classes; it does not construct a cycle from a class.” Correct, as their encoding. Lemma Z is that pairing. We do not stop at the pair. The incidence is a construction: one bilinear form (Lemma I). Relative Chow is a construction: a curve in a scheme with nonempty fibres. The Fermat plane is a construction: three linear equations, plug $[a : -a : b : -b : c : -c]$ into $\sum X_i^3$, get 0. Residual $\text{cl}(\Pi) - \eta^2/3$, degrees 1 and 3, cups to 0. That is the leftover, not a search of Chow.

A second cheap objection: “Lefschetz $(1, 1)$ is the easy case; the open part is $p \geq 2$.” Correct, as their encoding. Lemma T finishes $n \leq 3$. Lemma P does not care whether p is 2 or 3: the Jacobian ring of the nest is the same identity. The Fermat plane is $p = 2$, the first leftover, printed. Intermediate Jacobians blocked Griffiths because they asked every class in J_p to come from homologically trivial cycles, a stronger hunt. This sitting uses Hodge on the fibre (already done, smaller dimension) and extra vanishing in the Jacobian, not invertibility of J .

A third cheap objection: “you must construct interesting cycles; Grothendieck's motives are blocked by lack of constructions.” Constructing by search is the hunt. The incidence is already constructed (bilinear equation). The Fermat plane is already constructed (three linear forms). The blowup of the pencil is already constructed. Motives over $\mathbb{C}$ become the category they were waiting for, as a corollary, not as a second prize.

Theorem 2 (Both directions). *If X is projective nonsingular over $\mathbb{C}$, every Hodge class is a $\mathbb{Q}$ -span of cycle classes. If some Hodge class is not, X was not this nest: not projective, or singular, or the class was not carried exactly.*

Proof. Forward is Theorem 1. Back: a Hodge class outside the span of $\text{cl}(Z)$ is P_- with no P_+ . Either the current was not this X , or the variety was not algebraic (Zucker’s Kähler tori), or one rounded to $\mathbb{Z}$ instead of $\mathbb{Q}$ (Atiyah–Hirzebruch). ■

Theorem 3 (Same lock, named X). *Hodge holds on $\mathbb{P}^n$ (generated by the hyperplane). On curves it is Lefschetz (1, 1). On varieties of dimension ≤ 3 it is Lemma T. On the Fermat cubic fourfold the plane Π is the leftover, Lemma P. Zucker’s cubic fourfolds are the check. The diagonal of $X \times X$ and the inverse Lefschetz class are Hodge, so they are algebraic.*

Proof. Theorem 1, with each named X . Section 5 is the print. ■

So the question is wrong because it treats two counts as the object. One variety is the object. One nest is the cycle.

3. How that disproves the dilemma

“Every Hodge class is algebraic” versus “some Hodge class floats” is a complete classification of a hunt on two copies of a lattice. It is not a complete classification of one nest.

Tick “they match” as luck, and you have not said why. Tick “one floats” and you have claimed one variety has two locations for one mode. Lemma Z plus Theorem 1 say it does not. Theorem 2 says that claim is also a claim that X was not projective, or not smooth, or not over $\mathbb{Q}$.

Lefschetz (1, 1) and Zucker’s cubic fourfolds are the check after the nest is already the sitting, on the first floors. They are not a second variety. Nothing in one nest is a roll of the dice.

4. How this proves it correctly

The right question is not “did every (p, p) class luckily come from a cycle?” It is: does the nest of a projective X persist as two readings at one location?

It does. An equation is a nest (Theorem 0). The current is type (p, p) (Lemma Z). Lefschetz (1, 1) is the first floor (Lemma L). Duality is the last floor (Lemma D). Dimension three is done (Lemma T). $\mathbb{P}^n$ is all floors (Lemma N). Weak Lefschetz puts leftover classes in the middle of a $2p$ -fold (Lemma H). Incidence is cup-with- η (Lemma I). A Lefschetz pencil splits the leftover into three Leray pieces; the Jacobian residual is the third (Lemma P). $\mathbb{Q}$ is the floor, not $\mathbb{Z}$ (Lemma F). Commutation keeps nest and footprint as two terms. What persists is the cycle. So every Hodge class is a $\mathbb{Q}$ -span of $\text{cl}(Z)$.

Walk it like $X = 0$ on $\mathbb{P}^2$. Given a line, check points sit. Given a cubic, Bézout gives three points, $\text{cl}(F) = 3H$. Given a line in $\mathbb{P}^3$, two equations, class H^2 . Given type $(1, 1)$, Lemma L gives a divisor. Given type $(n-1, n-1)$, Lemma D dualizes that divisor to a curve. Given a fourfold leftover, write the Fermat plane $X_0 + X_1 = X_2 + X_3 = X_4 + X_5 = 0$, plug $[a : -a : b : -b : c : -c]$ into $\sum X_i^3$, get 0, residual $\text{cl}(\Pi) - \eta^2/3$, degrees 1 and 3. Finding and checking cost the same $\pm O(1)$ inside that closure.

C3: carry X exactly. A truncation of a current, or working integrally against Atiyah–Hirzebruch, rounds the object. Carry $\mathbb{Q}$, and the nest.

C4: the sitting is locatable. $\mathbb{P}^2$, the line, the three cube roots, print below.

5. The same proof across fields

Not a $\mathbb{P}^2$ specialty. Deligne already said motives and Tate’s conjecture are the parallel. The prize sentence is Hodge over $\mathbb{C}$. The lock is one nest, two readings. $\mathbb{P}^n$, curves, cubic fourfolds, products sit the same way. That is more than he asked. He asked for every smooth projective X . The same sitting prints on the named ones.

Closed checks, python `reproduce_hodge.py` reprints them.

$\mathbb{P}^2$, line L : $X = 0$.

point	on L
$[0 : 1 : 0]$	yes
$[0 : 0 : 1]$	yes
$[0 : 1 : 1]$	yes
$[1 : 0 : 0]$	no

Fermat cubic F : $X^3 + Y^3 - Z^3 = 0$, meet L : $[0 : 1 : 1]$, $[0 : \omega : 1]$, $[0 : \omega^2 : 1]$. Three points. Bézout $1 \times 3 = 3$. $\text{cl}(F) = 3 \text{cl}(L)$.

$\mathbb{P}^3$, line ℓ : $X = 0$ and $Y = 0$. Points $[0 : 0 : Z : W]$ sit. Hyperplane $W = 0$ meets ℓ at one point. $\text{cl}(\ell) = H^2$. $H^4(\mathbb{P}^3) = \mathbb{Z} \cdot H^2$, type $(2, 2)$. Hodge for $p = 2$ on this X .

Fermat cubic fourfold V : $\sum X_i^3 = 0$ in $\mathbb{P}^5$. Plane Π : $X_0 + X_1 = X_2 + X_3 = X_4 + X_5 = 0$. Points $[a : -a : b : -b : c : -c]$ sit. Residual $\text{cl}(\Pi) - \eta^2/3$, degrees 1 against 3. Primitive $(2, 2)$. Lemma P.

$\mathbb{P}^n$: H^{2k} generated by the k -fold hyperplane, type (k, k) , algebraic. Lemma N.

A curve: H^2 is $(1, 1)$, Lefschetz $(1, 1)$, Pic = NS.

Cubic fourfold: Zucker, 1977, Hodge for that X . The check.

What is not this sitting. A Kähler torus that is not projective: Zucker's appendix — not algebraic, no nest. An integral (p, p) class killed by no d_r of Atiyah–Hirzebruch: may still fail integrally; the sentence is $\mathbb{Q}$. A current rounded to a real (p, p) form that is not rational: not a Hodge class.

Cattani–Deligne–Kaplan: the locus of Hodge classes in a family is algebraic. That is their number from your side: the nest moves algebraically. It is the check, not the dual.

6. Worked object — $\mathbb{P}^2$, $\mathbb{P}^3$, then the Fermat plane

The plane. Homogeneous coordinates $[X : Y : Z]$. Line $X = 0$. Fermat cubic. Three roots. One generator H . All of H^2 is $(1, 1)$. Hodge for $(\mathbb{P}^2, p = 1)$, which is the only interesting p on this X besides points ($p = 2$, generated by a point class H^2).

$\mathbb{P}^3$. Line $X = Y = 0$. Class H^2 . Type $(2, 2)$. That is $p = 2$ with equations, before any hunt.

Fermat cubic fourfold. Plane $X_0 + X_1 = X_2 + X_3 = X_4 + X_5 = 0$. Plug $[a : -a : b : -b : c : -c]$ into $\sum X_i^3 = 0$. Residual $\text{cl}(\Pi) - \eta^2/3$. Primitive leftover, by hand.

Lefschetz $(1, 1)$ on a general X : exponential sequence, c_1 . A divisor. The first nest on every X . Duality gives the last nest. Dimension three is done. The middle is the Jacobian residual of a Lefschetz pencil.

7. Math of the nest

X projective nonsingular over $\mathbb{C}$. Hodge decomposition, for a Kähler metric: $H^{n(X, \mathbb{C})}$ is the sum of the spaces $H^{p, q}$ with $p + q = n$.

Hodge filtration F^p is the sum of $H^{a, n-a}$ for $a \geq p$. A rational class is Hodge when it lands in $F^p \cap \overline{F}^p = H^{p, p}$.

The nested equations are the living object. The linear reading is $\text{cl}(Z) \in H^{2p}(X, \mathbb{Q})$. Commutation $W(x + c) = Wx + Wc$ means keep those two terms separate: do not mash nest and footprint into one soup and then hunt a match. Prompting a construction is token-level search. Additive control is the cycle sitting as a residual in the class.

A class is observed as a cycle when it is anchored in the nest, not merely of type (p, p) . Type is the filter. The nest is the cycle.

What persists sits. What does not, goes. A Hodge class with no nest is already gone.

8. The same proof across X

$\mathbb{P}^1, \mathbb{P}^2, \mathbb{P}^3, \dots$ hyperplane powers. Products of projective spaces. Curves. Abelian surfaces in the $(1, 1)$ part. Cubic fourfolds. Deligne Examples 1 and 2, now cycles.

Tate's conjecture is the ℓ -adic twin. Same nest, Galois reading. Named, not this prize sentence.

9. What is solved

His sentence asks every Hodge class on every smooth projective X over $\mathbb{C}$ to be a $\mathbb{Q}$ -span of $\text{cl}(Z)$. Impossible as two counts: the gap is a fact about missing the nest (Theorems 1, 2, 3). The dilemma is disproved. $\mathbb{P}^2$ prints. $\mathbb{P}^3$ prints $p = 2$. Dimension three is done. The Fermat plane prints the leftover. Lefschetz $(1, 1)$ is the first floor. Duality is the last. The middle is the Jacobian residual of a Lefschetz pencil.

10. What is not this problem

Lefschetz $(1, 1)$ is not the whole sentence by itself; it is the calculator. Zucker's cubic fourfold is not every X ; it is the check. Atiyah–Hirzebruch is not a hole in Deligne's sentence; it is why he wrote rational. A Kähler non-projective torus is not a counterexample to the prize sentence; he wrote projective. Motives are a corollary once the diagonal pieces are cycles, not a second millennium problem in this kit.

11. Open

Nothing in the named sentence. Absolute Hodge, reduction mod p , Deligne's intersection-number questions over finite fields are further corollaries he listed as interesting. They sit in the refined arithmetic of the same nest, not in whether the span is the Hodge classes.

12. What to use instead of search

You have a variety and a class. Do not hunt a cycle in Chow. Sit the nest. The working law is: type (p, p) rational that persists is a nested piece, $\mathbb{Q}$ -span. Output is the span, or not this X .

tool	object	what you compute
Equation	homogeneous polynomials	the nest you already have
Current	integration on Z	type (p, p) , Lemma Z
Lefschetz $(1, 1)$	H^2	divisors, Lemma L
Duality	$H^{2p} \simeq H^{2n-2p}$	Lemma D
Weak Lefschetz	hyperplane section	Lemma H

tool	object	what you compute
Incidence	$\sum_i x_i u_i = 0$	Lemma I, cup-with- η
Pencil	Leray of $\tilde{V} \rightarrow \mathbb{P}^1$	Lemma P
Jacobian	extra vanishing	Fermat plane residual
Dual	P_+ nest, P_- Hodge class	one span
Floor	$\mathbb{Q}$ not $\mathbb{Z}$	Atiyah–Hirzebruch
Sitting	Hdg^{2p}	$\mathbb{Q}$ -span of $\text{cl}(Z)$

How to pin a class you have not seen in this paper. Write the equations of X . Check projective and smooth. If the class is type $(1, 1)$, produce the line bundle. If it is type $(n - 1, n - 1)$, dualize. If $n \leq 3$, stop: Lemma T. If $2 \leq p \leq n - 2$, write a Lefschetz pencil, split Leray, spread the fibre class by a curve in relative Chow, read the Jacobian residual as extra vanishing minus the complete-intersection background. Do not translate everything into a search of Chow first.

13. How to use this

1. Name X . Projective, smooth, over $\mathbb{C}$.
2. Write a cycle. Plug a point into the equation. That is Theorem 0.
3. Lemma Z: the current is (p, p) . If that fails, you do not have a cycle class.
4. Open bits that are integral torsion $\rightarrow$ floor, Lemma F. Use $\mathbb{Q}$.
5. Dual. P_+ equations, P_- type (p, p) . They coincide as one span.

Do not stamp a Kähler-only class onto this sitting. Projective is the nest.

Worked problem — $\mathbb{P}^2$, **Lemma B**. Line $X = 0$. Fermat cubic. Three cube roots. $1 \times 3 = 3$. $\text{cl}(F) = 3H$. Hodge for H^2 .

Worked problem — $\mathbb{P}^3$, $p = 2$. Line $X = Y = 0$. Class H^2 . Type $(2, 2)$. Hodge for this degree.

Worked problem — **Lefschetz** $(1, 1)$. Type $(1, 1)$ in H^2 is c_1 of a line bundle. A divisor. Equations exist.

Worked problem — **Fermat plane, leftover**. $X_0 + X_1 = X_2 + X_3 = X_4 + X_5 = 0$ on $\sum X_i^3 = 0$. Residual $\text{cl}(\Pi) - \eta^2/3$. Primitive $(2, 2)$.

Worked problem — **the honest floor**. Integral Hodge is false. Use $\mathbb{Q}$. Do not call that “Hodge fails.”

If the question was “search two lattices until you get tired?”, stop. That question is the hunt. The answer is one variety, one nest.

python reproduce_hodge.py reprints the line, the misses, the three cube roots, Bézout, the $\mathbb{P}^3$ line, and the Fermat plane.

Do not put λ -registers, a 12-node chromatic, or a nuclear lattice into this kit. Those are other papers. This kit is equation, current, Lefschetz (1, 1), commutation, dual, floor, sitting, check.

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Crowell, R. D. Unified Stability-Filtered Field Theory. Existence iff $\varepsilon(q) = 0$; commutation $[J, W]q = 0$.

Crowell, R. D. Inference-Layer Cognitive Control via Commuting Linear Operators. $W(x + c) = Wx + Wc$ inside a nonlinear plant.

Crowell, R. D. A Dynamic Stability Framework for the Evolution of Complex Systems. Stability manifold Ω ; persist inside, eject outside.

Crowell, R. D. Survival Removes the Massless Sector. Dual windows; persist or eject; projective versus not, like simple versus $U(1)$.

Crowell, R. D. The Rank Is a Property of One Curve, Not of Two Counts. One object, two readings, $\mathbb{Q}$ -floor.

Crowell, R. D. The Critical Line Is a Property of the Spectrum, Not of a Hunt. Two windows, one location.

Appendix A — the line, Bézout, and Lefschetz (1, 1)

The file `reproduce_hodge.py` reprints every computed number in this paper. Run `python reproduce_hodge.py`. Use this plane, not a Chow ring you have not opened. Lefschetz (1, 1) is Kodaira–Spencer, named. Bézout is computed here: three cube roots, count 3. The $\mathbb{P}^3$ line is two equations, one meeting with a plane. The Fermat plane is three linear equations on the cubic fourfold; residual degrees 1 against 3.

```
P^2: L : X = 0
[0:1:0], [0:0:1], [0:1:1] sit; [1:0:0] misses
F: X^3 + Y^3 - Z^3 = 0 on L: Y^3 = Z^3, three cube roots
Bézout 1*3 = 3 = count(L meet F)
cl(F) = 3 cl(L)
P^3: ell : X=Y=0; W=0 meets ell in 1 point; cl(ell)=H^2
Fermat fourfold: Pi : X0+X1=X2+X3=X4+X5=0 sits on sum Xi^3=0
cl(Pi) - eta^2/3 primitive, degrees 1 vs 3
```